Paper 1 Experimental Results

Cybersecurity Data Engineering Framework

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1 Experimental Environment

The experiments were conducted using Python, Jupyter Notebook, Pandas, NumPy, PyTorch, and Hugging Face Transformers.

Three benchmark datasets were integrated:

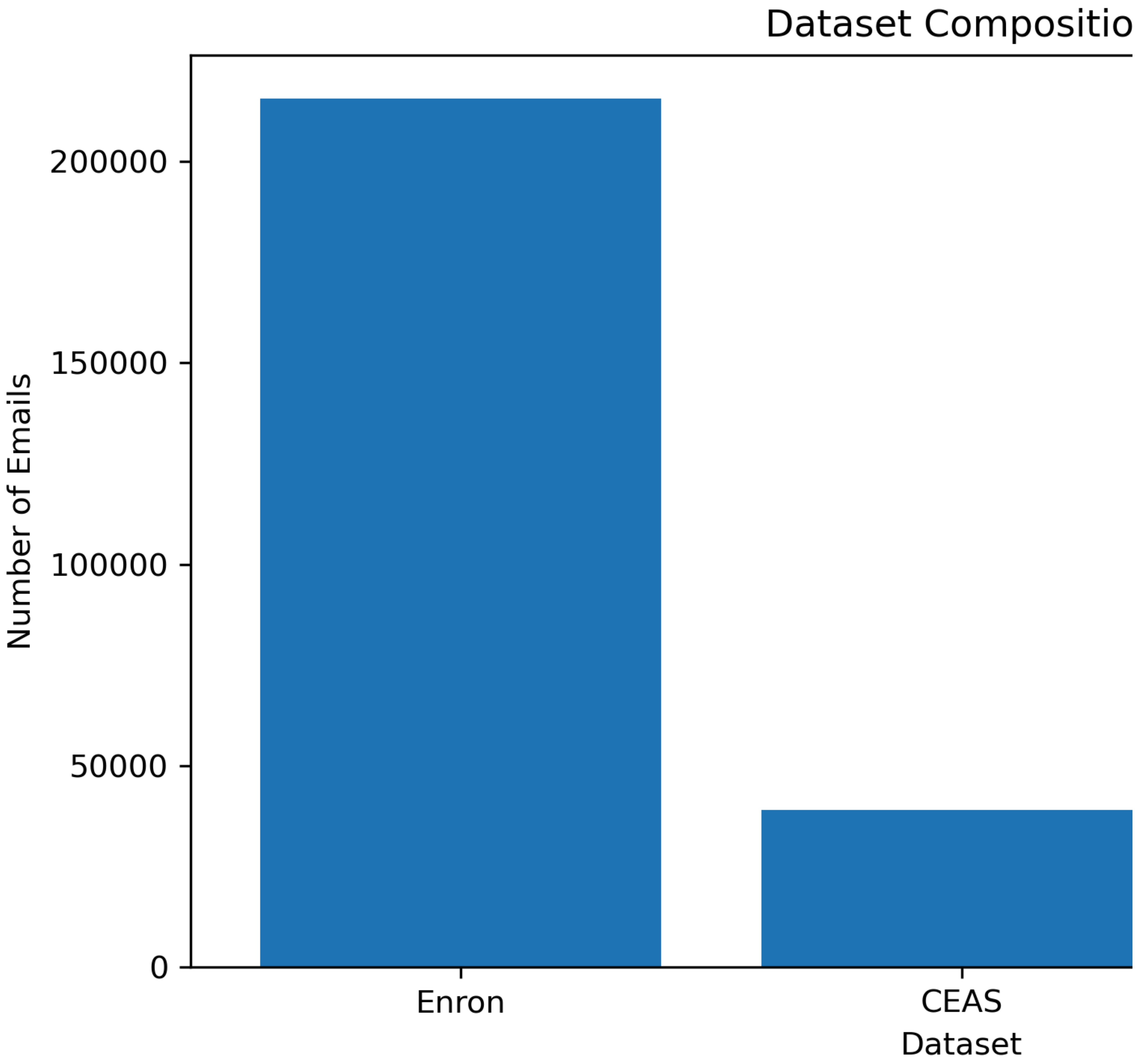
- CEAS 2008
- Enron
- SpamAssassin

2 Dataset Composition

Table 1 Dataset Composition

Dataset	Emails	Percentage
Enron	215638	83.37
CEAS	39039	15.09
SpamAssassin	3979	1.54

Figure 1 Dataset Composition



3 Data Quality Evaluation

Table 2 Data Quality Summary

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Figure 2 Duplicate Reduction

Figure 2 Duplicate Reduction
Figure 2 Duplicate Reduction

Figure 3 Missing Values

Figure 3 Missing Values # 4 Cleaning Evaluation

Table 3 Cleaning Evaluation

Missing: table_03_cleaning_evaluation.csv # 5 Transformer Tokenization Evaluation

Table 4 Tokenization Statistics

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Figure 4 Token Length Distribution

Figure 4 Token Length Distribution # 6 Computational Performance

Table 5 Computational Performance

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Table 6 Master Evaluation Summary

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Figure 5 Proposed Cybersecurity Data Engineering Framework

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8 Discussion

The quantitative results demonstrate the effectiveness of the proposed cybersecurity data engineering framework for preparing heterogeneous benchmark email datasets for Transformer-based phishing detection.

The framework successfully standardized multiple datasets, removed redundant observations, cleaned textual noise while preserving security-relevant information, and produced a Transformer-compatible corpus suitable for contextual embedding generation.

9 Conclusion

The experimental evaluation confirms that the proposed framework provides a robust, reproducible, and scalable preprocessing pipeline for phishing email research.

The resulting cleaned and tokenized corpus provides the foundation for Student 2’s contextual embedding generation.